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$$\int \frac{dx}{(x+1)\sqrt{x}}$$

$$\text{Stel } t = \sqrt{x} \Rightarrow dx = 2t dt$$

$$= \int \frac{2t}{(t^2+1)t} dt$$

$$= \int \frac{2}{t^2+1} dt$$

$$= 2 \int \frac{dt}{t^2+1}$$

$$= 2 \operatorname{Bgtan}(t) + C$$

$$= 2 \operatorname{Bgtan}(\sqrt{x}) + C$$

References